PAVAN SRINIVAS DOMA

Data Engineer II
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Data Engineer with over a year of hands-on experience at the University at Buffalo and a Master's in Engineering Science. Skilled in SQL, Python and data modeling, with academic projects in retrieval-augmented conversational AI and credit scoring & customer segmentation. Raised asset utilization efficiency by 25% after refactoring SQL queries and redesigning the ETL schema.

SKILLS

Languages: Python, SQL
Data Engineering: Data Pipelines, Window Functions, Data Modeling, Data Quality, Query Optimization, ETL
Data & Analytics: Matplotlib, NumPy, Pandas, KPI Reporting, Data Analytics
Cloud & DevOps: AWS, Lambda, ECS, EC2

PROFESSIONAL EXPERIENCE

Data Engineer - Lead Buffalo, USA
University at Buffalo Sep 2024 - Dec 2025

- Improved ETL pipeline performance through SQL query refactoring and schema redesign, supporting real-time operational decisions that raised asset utilization efficiency by 25%.
- Standardized and documented ETL pipelines for stock tracking and allocation, adding data quality gates and error handling that cut data ingestion errors by 30%; wrote runbooks that helped onboard new engineers faster.
- Built a production resource allocation pipeline with SQL window functions and time-series decomposition, working with stakeholders to validate the business logic from extraction to the insights shared with leadership.
- Designed and maintained a normalized schema for people, inventory and transactions with the operations team, using validation rules and data quality checks to keep transactional integrity at 100% for reliable reporting.
- Built real-time KPI dashboards in advanced SQL (window functions, CTEs, aggregations) and optimized the queries behind them, cutting query runtime by 40%.

ACADEMIC PROJECTS

Retrieval-Augmented Conversational AI Systems

- Designed a retrieval-augmented generation (RAG) system and validated its semantic relevance at 97.5% accuracy with ML engineers; containerized it on AWS ECS with monitoring.
- Implemented context engineering, conversation memory and structured output validation so the assistant gave consistent, context-aware answers.

Credit Scoring & Customer Segmentation

- Compared 5 model approaches in MLflow with senior engineers and checked the business logic against data quality trade-offs.
- Developed a credit risk scoring model and defined its feature engineering strategy with a cross-functional team.

EDUCATION

University at Buffalo, SUNY Buffalo, USA
Master's in Engineering Science Jan 2026

S.R.M Institute of Science and Technology, SRM University Chennai, India
Bachelor of Technology in Artificial Intelligence May 2024

CERTIFICATIONS

- Claude with Amazon Bedrock (Generative AI & LLM Application Development) Jul 2026
- AWS Cloud Practitioner (Cloud platform deployment & AI services) Jan 2026